Check-In Challenge Review Packet

5th Grade Science (NC) and Equivalent Fractions (NC.4.NF.1)

Student name: _____ **Date:** _____

Teacher: _____ **Deck played:** _____

My Standards Chart

After you play the game, write your score for each standard. Shade the bar to match your percent, then check one box. Any standard below 60% is the one you need more help with.

Standard	Score	0% 25% 50% 75% 100%	Needs help	Almost there	On track
SCIENCE					
Matter	___ / 8		☐	☐	☐
Forces & Motion	___ / 6		☐	☐	☐
Ecosystems	___ / 6		☐	☐	☐
Cells & Body Systems	___ / 5		☐	☐	☐
Weather	___ / 5		☐	☐	☐
FRACTIONS					
NC.4.NF.1 · Identify equivalent	___ / 9		☐	☐	☐
NC.4.NF.1 · Find the missing number	___ / 7		☐	☐	☐
NC.4.NF.1 · Use fraction models	___ / 5		☐	☐	☐
NC.4.NF.1 · Word problems	___ / 5		☐	☐	☐
NC.4.NF.1 · Explain & prove	___ / 4		☐	☐	☐

I need more help with: _____

My goal for next time: _____

Family signature: _____

Part 1 · 5th Grade Science

Circle the best answer. Answers and explanations are at the back.

Matter

1. A student pours 200 g of water into a cup and stirs in 20 g of salt until it dissolves. What is the mass of the salt water?

- **A** 180 g
- **B** 200 g
- **C** 220 g
- **D** 240 g

Matter

2. Which of these is a chemical change?

- **A** An ice cube melting
- **B** An iron nail rusting
- **C** Paper being cut in half
- **D** Sugar dissolving in tea

Matter

3. Baking soda and vinegar are sealed in a plastic bag. The bag has a mass of 150 g. The mixture fizzes and the bag puffs up. What is the mass of the sealed bag now?

- **A** Less than 150 g
- **B** Exactly 150 g
- **C** More than 150 g
- **D** It cannot be measured

Matter

4. A 30 g ice cube melts completely in a closed container. What is the mass of the water?

- **A** 15 g
- **B** 25 g
- **C** 30 g
- **D** 35 g

Matter

5. Which mixture could be separated most easily using a magnet?

- **A** Salt and water
- **B** Sand and iron filings
- **C** Sugar and flour
- **D** Rice and beans

Matter

6. Which observation is the BEST evidence that a new substance formed?

- **A** Water turns to steam when heated
- **B** Clay is rolled into a ball
- **C** Two liquids are mixed, bubbles form, and the cup gets warm
- **D** Juice is poured into a taller glass

Matter

7. Which property tells how much space an object takes up?

- **A** Mass
- **B** Volume
- **C** Color
- **D** Temperature

Matter

8. Which substance will dissolve in water?

- **A** Sand
- **B** Pebbles
- **C** Cooking oil
- **D** Sugar

Forces & Motion

9. A ball is thrown straight up into the air. Why does it fall back to the ground?

- **A** Friction pulls it down
- **B** Gravity pulls it toward Earth
- **C** Air pushes it down
- **D** It runs out of energy and stops

Forces & Motion

10. A car travels 120 miles in 2 hours at a steady speed. What is its speed?

- **A** 30 miles per hour
- **B** 60 miles per hour
- **C** 118 miles per hour
- **D** 240 miles per hour

Forces & Motion

11. Two students push identical toy cars on the same floor. Kim pushes much harder than Leo. What most likely happens?

- **A** Both cars travel the same distance
- **B** Leo's car travels farther
- **C** Kim's car moves faster and farther
- **D** Neither car moves

Forces & Motion

12. A toy car rolls off a ramp onto four different surfaces. On which surface will it stop the soonest?

- **A** Smooth tile
- **B** Ice
- **C** Wood floor
- **D** Thick carpet

Forces & Motion

13. Two teams pull on a rope with equal force in a tug-of-war. What happens to the rope?

- **A** It moves toward the stronger team
- **B** It moves up into the air
- **C** It does not move
- **D** It speeds up

Forces & Motion

14. A soccer ball is sitting still on the grass. What will make it start moving?

- **A** An unbalanced force, like a kick
- **B** Balanced forces acting on it
- **C** Gravity alone
- **D** Nothing can make it move

Ecosystems

15. In the food chain grass → rabbit → fox, where does the energy first come from?

- **A** The soil
- **B** The sun
- **C** The rabbit
- **D** The fox

Ecosystems

16. What is the main job of decomposers like mushrooms and bacteria?

- **A** Make food from sunlight
- **B** Hunt other animals
- **C** Break down dead things and return nutrients to the soil
- **D** Pollinate flowers

Ecosystems

17. In a field, snakes eat mice. If all the snakes were removed, what would most likely happen to the mice at first?

- **A** The mouse population would increase
- **B** The mouse population would decrease
- **C** The mice would become producers
- **D** Nothing would change

Ecosystems

18. Which organism is a producer?

- **A** Hawk
- **B** Oak tree
- **C** Mushroom
- **D** Deer

Ecosystems

19. In a food web, which way do the arrows point?

- **A** From the predator to the prey
- **B** From the animal to the sun
- **C** From bigger animals to smaller animals
- **D** From the organism eaten to the organism that eats it

Ecosystems

20. A deer eats only plants. What kind of consumer is it?

- **A** Herbivore
- **B** Carnivore
- **C** Omnivore
- **D** Decomposer

Cells & Body Systems

21. Which part of a cell controls the cell's activities?

- **A** Cell membrane
- **B** Cytoplasm
- **C** Nucleus
- **D** Cell wall

Cells & Body Systems

22. Which two body systems work together to get oxygen to every part of the body?

- **A** Skeletal and muscular
- **B** Digestive and nervous
- **C** Respiratory and circulatory
- **D** Nervous and skeletal

Cells & Body Systems

23. How do muscles help you bend your arm?

- **A** Muscles push on bones
- **B** Muscles pull on bones
- **C** Bones pull on muscles
- **D** Nerves pull on bones

Cells & Body Systems

24. Which body system breaks food down into nutrients the body can use?

- **A** Circulatory system
- **B** Skeletal system
- **C** Respiratory system
- **D** Digestive system

Cells & Body Systems

25. You touch a hot stove and pull your hand away fast. Which system sent the message to move?

- **A** Nervous system
- **B** Digestive system
- **C** Respiratory system
- **D** Skeletal system

Weather

26. A cold front is moving into central North Carolina. What weather should people most likely expect?

- **A** Clear skies for a week
- **B** Thunderstorms, then cooler temperatures
- **C** Warmer, more humid air
- **D** No change in the weather

Weather

27. Water vapor rises, cools, and forms clouds. What is this step of the water cycle called?

- **A** Evaporation
- **B** Precipitation
- **C** Condensation
- **D** Runoff

Weather

28. A barometer shows the air pressure is falling quickly. What does this usually mean?

- **A** Clear, sunny weather is coming
- **B** Cloudy, stormy weather is likely
- **C** The temperature will stay the same
- **D** It is about to get windless

Weather

29. Which tool measures wind speed?

- **A** Thermometer
- **B** Rain gauge
- **C** Barometer
- **D** Anemometer

Weather

30. Tall, dark cumulonimbus clouds are building in the afternoon. What weather do they usually bring?

- **A** Thunderstorms
- **B** Clear skies
- **C** Light fog
- **D** Fair weather all day

Part 2 · Equivalent Fractions (NC.4.NF.1)

Circle the best answer. Answers and explanations are at the back.

NC.4.NF.1 · Identify equivalent

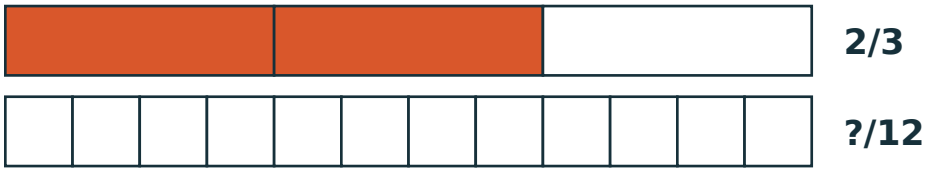
31. Which fraction is equivalent to 1/2?



A 2/3
B 3/5
C 4/8
D 5/12

NC.4.NF.1 · Find the missing number

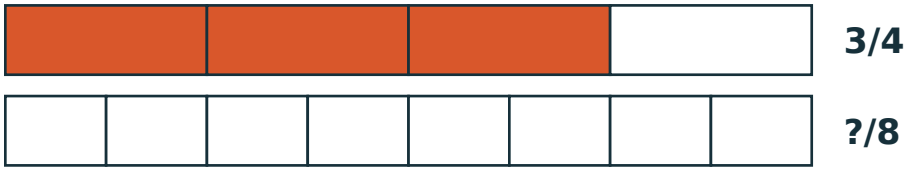
32. 2/3 = ?/12. What number goes in the box?



A 6
B 8
C 9
D 10

NC.4.NF.1 · Use fraction models

33. The top bar shows 3/4 shaded. The bottom bar is the same size but has 8 equal parts. How many parts should be shaded so both show the same amount?



A 3
B 4
C 6
D 7

NC.4.NF.1 · Identify equivalent

34. Which fraction is NOT equivalent to 2/5?

A 4/10
B 6/15
C 4/15
D 8/20

NC.4.NF.1 · Word problems

35. Maria ate 6/10 of a pizza. Ty ate 3/5 of a pizza the same size. Who ate more?



6/10

3/5

A Maria
B Ty
C They ate the same amount
D There is not enough information

NC.4.NF.1 · Explain & prove

36. Which shows the correct way to make a fraction equivalent to 3/4?

A (3 + 2) / (4 + 2) = 5/6
B (3 × 2) / (4 × 2) = 6/8
C (3 × 2) / 4 = 6/4
D 3 / (4 × 2) = 3/8

NC.4.NF.1 · Identify equivalent

37. Which fraction is equivalent to 9/12?



9/12

A 3/4
B 2/3
C 9/10
D 1/3

NC.4.NF.1 · Use fraction models

38. A point on a number line is at 2/6. Which fraction names the same point?



2/6

A 1/2
B 1/3
C 2/3
D 1/6

NC.4.NF.1 · Identify equivalent

39. Which pair of fractions is equivalent?

A 5/6 and 10/12
B 3/8 and 4/9
C 2/4 and 3/8
D 1/5 and 2/5

NC.4.NF.1 · Find the missing number

40. 4/10 = ?/100. What number goes in the box?

A 4
B 14
C 40
D 400

NC.4.NF.1 · Word problems

41. A recipe needs 1/4 cup of milk. Ana only has a 1/8 cup scoop. How many scoops does she need?



1/4

A 1
B 2
C 4
D 8

NC.4.NF.1 · Find the missing number

42. 3/? = 12/16. What number goes in the box?

A 3
B 4
C 5
D 8

NC.4.NF.1 · Use fraction models

43. A rectangle is cut into 9 equal parts. How many parts must be shaded to show 2/3?



2/3

A 2
B 3
C 5
D 6

NC.4.NF.1 · Explain & prove

44. Jamal says 2/3 = 4/5 because he added 2 to the top and bottom. Why is he wrong?

A He should have added 3
B Adding does not keep the fractions equal; he must multiply both by the same number
C Fractions can never be equivalent
D He should have subtracted 2

NC.4.NF.1 · Identify equivalent

45. Which fraction is equivalent to 1 whole?

A 1/8
B 7/8
C 8/8
D 8/1

NC.4.NF.1 · Word problems

46. A garden has 12 rows. 8 rows are corn. Which fraction is equivalent to the part that is corn?



8/12

A 1/2
B 2/3
C 3/4
D 4/5

NC.4.NF.1 · Identify equivalent

47. Which list shows fractions that are ALL equivalent to 1/3?

A 2/6, 3/9, 4/12
B 2/3, 3/4, 4/5
C 1/6, 1/9, 1/12
D 3/1, 6/2, 9/3

NC.4.NF.1 · Use fraction models

48. The bar shows 5/10 shaded. Which fraction is equivalent?



5/10

A 1/5
B 1/2
C 5/5
D 2/5

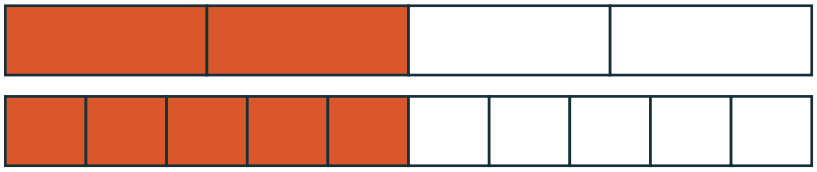
NC.4.NF.1 · Find the missing number

49. 3/8 = ?/24. What number goes in the box?

A 6
B 9
C 12
D 16

NC.4.NF.1 · Word problems

50. Lena read 2/4 of her book. Max read 5/10 of the same book. Which statement is true?



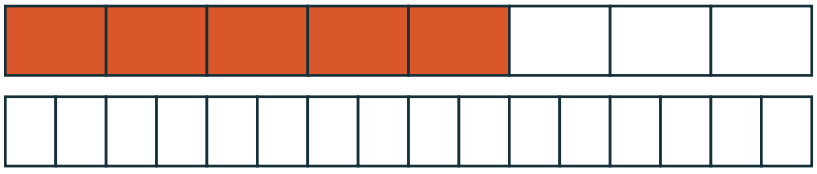
2/4

5/10

A Lena read more
B Max read more
C They both read 1/2 of the book
D They both read 1/4 of the book

NC.4.NF.1 · Find the missing number

51. 5/8 = ?/16. What number goes in the box?



5/8

?/16

A 5
B 8
C 10
D 13

NC.4.NF.1 · Identify equivalent

52. Which fraction is equivalent to 10/12?



10/12

A 1/2
B 2/3
C 3/4
D 5/6

NC.4.NF.1 · Find the missing number

53. 1/5 = ?/100. What number goes in the box?

A 5
B 20
C 50
D 95

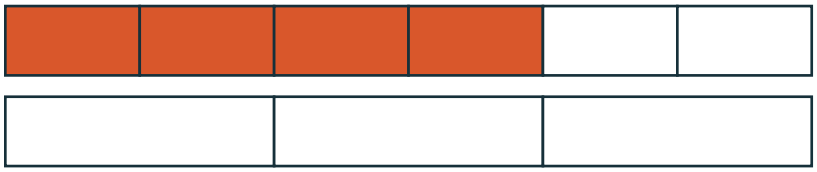
NC.4.NF.1 · Identify equivalent

54. Which number sentence is true?

A 2/3 = 4/6
B 1/4 = 2/6
C 3/5 = 5/7
D 4/8 = 1/4

NC.4.NF.1 · Use fraction models

55. The top bar shows 4/6 shaded. The bottom bar is the same size with 3 equal parts. How many parts should be shaded so both show the same amount?



4/6

?/3

A 1
B 2
C 3
D 4

NC.4.NF.1 · Explain & prove

56. Ella says 3/6 and 4/8 are both equal to 1/2. Is she correct?

A Yes, because 3 is half of 6 and 4 is half of 8
B No, because 4/8 has bigger numbers
C No, because the denominators are different
D Only 3/6 equals 1/2

NC.4.NF.1 · Find the missing number

57. 5/? = 15/18. What number goes in the box?

A 3
B 5
C 6
D 8

NC.4.NF.1 · Word problems

58. 8 out of 10 students brought a pencil. Which fraction is equivalent to 8/10?



8/10

A 2/5
B 3/5
C 8/5
D 4/5

NC.4.NF.1 · Explain & prove

59. Which shows how to prove that 2/5 = 6/15?

A 2/5 + 4/10 = 6/15
B (2 × 3) / (5 × 3) = 6/15
C (2 + 4) / (5 + 10) = 6/15
D (2 × 3) / 5 = 6/15

NC.4.NF.1 · Identify equivalent

60. Which fraction is equivalent to 3/12?



3/12

A 1/4
B 1/3
C 3/4
D 1/12

Answer Key and Explanations

✓ marks the correct answer. Every choice explains why it is right or wrong.

1. Answer: C **(Matter)**

✗ **A** 180 g: This subtracts the salt. The salt didn't leave the cup, so its mass is still counted.
✗ **B** 200 g: This counts only the water. Dissolved salt is invisible, but it is still there and still has mass.
✓ **C** 220 g: Mass is conserved: 200 g + 20 g = 220 g. Dissolving doesn't destroy any matter.
✗ **D** 240 g: This adds 20 g twice. Nothing new was added, so the total is 200 g + 20 g.

2. Answer: B **(Matter)**

✗ **A** An ice cube melting: Melting is a physical change. The water is still water, just in liquid form.
✓ **B** An iron nail rusting: Rust is a new substance formed when iron reacts with oxygen and water. That makes it a chemical change.
✗ **C** Paper being cut in half: Cutting changes size and shape only. Each piece is still paper, so this is a physical change.
✗ **D** Sugar dissolving in tea: Dissolving is a physical change. The sugar is still sugar and could be recovered by evaporating the water.

3. Answer: B **(Matter)**

✗ **A** Less than 150 g: Mass would go down only if gas escaped. The bag is sealed, so the gas is trapped inside.
✓ **B** Exactly 150 g: In a closed system, mass is conserved. The gas is new, but it is made from matter already in the bag.
✗ **C** More than 150 g: No matter entered the sealed bag. Puffing up means more volume, not more mass.
✗ **D** It cannot be measured: You can measure it on a scale. The sealed bag will still read 150 g.

4. Answer: C **(Matter)**

✗ **A** 15 g: Melting doesn't cut matter in half. Changing state doesn't change mass.
✗ **B** 25 g: No water escaped from the closed container, so no mass was lost.
✓ **C** 30 g: Melting is a physical change. The water has the same mass as the ice did.
✗ **D** 35 g: Nothing was added. Liquid water has the same mass as the ice it came from.

5. Answer: B **(Matter)**

✗ **A** Salt and water: Salt is not magnetic. Evaporation would separate this mixture.
✓ **B** Sand and iron filings: Iron is magnetic. A magnet pulls the filings out and leaves the sand behind.
✗ **C** Sugar and flour: Neither is magnetic. A magnet would not pull either one out.
✗ **D** Rice and beans: Neither is magnetic. You could sort these by hand or with a sieve.

6. Answer: C **(Matter)**

✗ **A** Water turns to steam when heated: Steam is still water in gas form. This is a physical change.
✗ **B** Clay is rolled into a ball: Only the shape changed. The clay is still clay.
✓ **C** Two liquids are mixed, bubbles form, and the cup gets warm: A gas forming and a temperature change are signs of a chemical change, which makes a new substance.
✗ **D** Juice is poured into a taller glass: The juice only changed containers. Its substance didn't change.

7. Answer: B **(Matter)**

✗ **A** Mass: Mass is the amount of matter in an object, not the space it fills.
✓ **B** Volume: Volume is the amount of space an object takes up.
✗ **C** Color: Color is a property you can see. It doesn't tell how much space something fills.
✗ **D** Temperature: Temperature tells how hot or cold something is, not its size.

8. Answer: D **(Matter)**

✗ **A** Sand: Sand sinks to the bottom and stays visible. It doesn't dissolve.
✗ **B** Pebbles: Pebbles don't break apart and spread through the water, so they don't dissolve.
✗ **C** Cooking oil: Oil floats on top of water in a layer. It doesn't dissolve.
✓ **D** Sugar: Sugar spreads evenly through the water and disappears from view. It dissolves and forms a solution.

9. Answer: B (Forces & Motion)

✗ **A** Friction pulls it down: Friction slows objects that rub together. It doesn't pull things toward Earth.
✓ **B** Gravity pulls it toward Earth: Gravity is the force that pulls objects toward the center of Earth.
✗ **C** Air pushes it down: Air resistance slows the ball, but it doesn't cause the fall.
✗ **D** It runs out of energy and stops: Objects don't just stop. A force, gravity, changes the ball's motion and pulls it down.

10. Answer: B (Forces & Motion)

✗ **A** 30 miles per hour: This divides by 4 instead of 2. Speed = 120 ÷ 2.
✓ **B** 60 miles per hour: Speed = distance ÷ time. 120 miles ÷ 2 hours = 60 miles per hour.
✗ **C** 118 miles per hour: This subtracts instead of dividing. Speed means distance divided by time.
✗ **D** 240 miles per hour: This multiplies instead of dividing. Speed = distance ÷ time.

11. Answer: C (Forces & Motion)

✗ **A** Both cars travel the same distance: The forces are different, so the changes in motion will be different too.
✗ **B** Leo's car travels farther: Leo used the smaller force. A smaller push makes a smaller change in motion.
✓ **C** Kim's car moves faster and farther: A greater force causes a greater change in motion, so Kim's car speeds up more and goes farther.
✗ **D** Neither car moves: Both students push the cars, so both cars move.

12. Answer: D (Forces & Motion)

✗ **A** Smooth tile: Tile is smooth, so it has low friction. The car would roll a long way.
✗ **B** Ice: Ice has very little friction. The car would roll the farthest.
✗ **C** Wood floor: Wood is fairly smooth. The car would roll farther than on carpet.
✓ **D** Thick carpet: Carpet is rough, so it creates the most friction and slows the car fastest.

13. Answer: C (Forces & Motion)

✗ **A** It moves toward the stronger team: Neither team is stronger here. The forces are equal.
✗ **B** It moves up into the air: The pulls are sideways. Nothing is pulling the rope upward.
✓ **C** It does not move: Equal forces in opposite directions are balanced. Balanced forces don't change motion.
✗ **D** It speeds up: Only unbalanced forces make an object speed up.

14. Answer: A (Forces & Motion)

✓ **A** An unbalanced force, like a kick: An unbalanced force changes an object's motion. A kick starts the ball moving.
✗ **B** Balanced forces acting on it: Balanced forces cancel out. The ball would stay still.
✗ **C** Gravity alone: Gravity pulls the ball down into the ground. It doesn't roll the ball across flat grass.
✗ **D** Nothing can make it move: Any unbalanced force, like a push or a kick, can start it moving.

15. Answer: B (Ecosystems)

✗ **A** The soil: Soil gives plants water and nutrients, but not the energy they use to make food.
✓ **B** The sun: Grass uses sunlight to make food. That energy then passes to the rabbit and the fox.
✗ **C** The rabbit: The rabbit gets its energy by eating grass, so it isn't the first source.
✗ **D** The fox: The fox is at the end of the chain. It gets energy from the rabbit.

16. Answer: C (Ecosystems)

✗ **A** Make food from sunlight: That is what producers, like plants, do.
✗ **B** Hunt other animals: That is what predators do.
✓ **C** Break down dead things and return nutrients to the soil: Decomposers recycle nutrients from dead plants and animals so producers can use them again.
✗ **D** Pollinate flowers: Bees and other pollinators do that job, not decomposers.

17. Answer: A (Ecosystems)

✓ **A** The mouse population would increase: With no predators eating them, more mice survive, so their population grows at first.

✗ **B** The mouse population would decrease: The mice lost a predator, not a food source, so their numbers would go up.

✗ **C** The mice would become producers: Mice are consumers. They can't make their own food from sunlight.

✗ **D** Nothing would change: Removing a predator affects the whole food web, starting with its prey.

18. Answer: B (Ecosystems)

✗ **A** Hawk: A hawk eats other animals. It is a consumer.

✓ **B** Oak tree: Plants make their own food using sunlight, so an oak tree is a producer.

✗ **C** Mushroom: A mushroom is a decomposer. It breaks down dead material.

✗ **D** Deer: A deer eats plants. It is a consumer.

19. Answer: D (Ecosystems)

✗ **A** From the predator to the prey: This is backwards. Energy moves from the prey to the predator that eats it.

✗ **B** From the animal to the sun: The sun gives energy to producers. Energy doesn't flow back to the sun.

✗ **C** From bigger animals to smaller animals: Arrows show who eats whom, not size.

✓ **D** From the organism eaten to the organism that eats it: Arrows show the flow of energy, from the food to the eater.

20. Answer: A (Ecosystems)

✓ **A** Herbivore: Herbivores eat only plants, so a deer is a herbivore.

✗ **B** Carnivore: Carnivores eat only other animals.

✗ **C** Omnivore: Omnivores eat both plants and animals.

✗ **D** Decomposer: Decomposers break down dead things. A deer eats living plants.

21. Answer: C (Cells & Body Systems)

✗ **A** Cell membrane: The membrane controls what goes in and out of the cell, not all of its activities.

✗ **B** Cytoplasm: Cytoplasm is the jelly-like fluid that holds the cell parts.

✓ **C** Nucleus: The nucleus is the control center of the cell.

✗ **D** Cell wall: The cell wall gives plant cells support. Animal cells don't have one.

22. Answer: C (Cells & Body Systems)

✗ **A** Skeletal and muscular: These systems work together to move the body, not to deliver oxygen.

✗ **B** Digestive and nervous: Digestion breaks down food, and nerves send signals. Neither one carries oxygen.

✓ **C** Respiratory and circulatory: The lungs take in oxygen, and the heart and blood carry it through the body.

✗ **D** Nervous and skeletal: These systems control and support the body. They don't move oxygen.

23. Answer: B (Cells & Body Systems)

✗ **A** Muscles push on bones: Muscles can only pull. They contract, or shorten, and can't push.

✓ **B** Muscles pull on bones: Muscles contract and pull on bones to make them move.

✗ **C** Bones pull on muscles: Bones can't move by themselves. Muscles move them.

✗ **D** Nerves pull on bones: Nerves send the signal, but muscles do the pulling.

24. Answer: D (Cells & Body Systems)

✗ **A** Circulatory system: The circulatory system carries nutrients around the body after food is broken down.

✗ **B** Skeletal system: The skeletal system supports and protects the body.

✗ **C** Respiratory system: The respiratory system takes in oxygen and releases carbon dioxide.

✓ **D** Digestive system: The stomach and intestines break food down into nutrients.

25. Answer: A **(Cells & Body Systems)**

✓ **A** Nervous system: Nerves and the brain sense the heat and send a signal to your muscles.
✗ **B** Digestive system: The digestive system breaks down food. It doesn't sense heat.
✗ **C** Respiratory system: The respiratory system handles breathing, not reactions.
✗ **D** Skeletal system: Bones support your body, but they don't send signals.

26. Answer: B **(Weather)**

✗ **A** Clear skies for a week: Fronts bring changing weather, not a long calm stretch.
✓ **B** Thunderstorms, then cooler temperatures: A cold front pushes warm air up fast, which often causes storms. Then cooler air moves in.
✗ **C** Warmer, more humid air: That describes a warm front, not a cold front.
✗ **D** No change in the weather: When air masses meet at a front, the weather changes.

27. Answer: C **(Weather)**

✗ **A** Evaporation: Evaporation is liquid water turning into vapor. That happens before clouds form.
✗ **B** Precipitation: Precipitation is water falling as rain, snow, sleet, or hail.
✓ **C** Condensation: Condensation is water vapor cooling into tiny liquid drops, which form clouds.
✗ **D** Runoff: Runoff is water flowing over land into streams and rivers.

28. Answer: B **(Weather)**

✗ **A** Clear, sunny weather is coming: Rising pressure usually means clear weather.
✓ **B** Cloudy, stormy weather is likely: Falling air pressure usually means clouds and precipitation are on the way.
✗ **C** The temperature will stay the same: A barometer measures air pressure, not temperature.
✗ **D** It is about to get windless: Falling pressure often brings more wind, not less.

29. Answer: D **(Weather)**

✗ **A** Thermometer: A thermometer measures temperature.
✗ **B** Rain gauge: A rain gauge measures how much rain falls.
✗ **C** Barometer: A barometer measures air pressure.
✓ **D** Anemometer: An anemometer has spinning cups that measure wind speed.

30. Answer: A **(Weather)**

✓ **A** Thunderstorms: Cumulonimbus clouds are tall storm clouds that bring thunder, lightning, and heavy rain.
✗ **B** Clear skies: Clear skies have few or no clouds.
✗ **C** Light fog: Fog is a cloud that forms near the ground, not a tall storm cloud.
✗ **D** Fair weather all day: Puffy white cumulus clouds mean fair weather. Cumulonimbus clouds mean storms.

31. Answer: C **(NC.4.NF.1 · Identify equivalent)**

✗ **A** 2/3: 2/3 is more than half. Half of 3 parts would be 1½ parts, not 2.
✗ **B** 3/5: 3/5 is more than half. Half of 5 parts would be 2½ parts.
✓ **C** 4/8: 1/2 × 4/4 = 4/8. Four out of eight equal parts is exactly half.
✗ **D** 5/12: 5/12 is less than half. Half of 12 is 6, so 6/12 would equal 1/2.

32. Answer: B **(NC.4.NF.1 · Find the missing number)**

✗ **A** 6: This takes half of 12. Multiply instead: 3 × 4 = 12, so 2 × 4 = 8.
✓ **B** 8: 3 × 4 = 12, so multiply the numerator by 4 too: 2 × 4 = 8.
✗ **C** 9: 9/12 = 3/4, which is more than 2/3.
✗ **D** 10: This adds 8 to the numerator. Use the same multiplier as the denominator.

33. Answer: C **(NC.4.NF.1 · Use fraction models)**

✗ **A** 3: 3/8 is less than 3/4, because eighths are smaller pieces than fourths.
✗ **B** 4: 4/8 is only half the bar. 3/4 is more than half.
✓ **C** 6: 4 × 2 = 8, so 3 × 2 = 6. 3/4 = 6/8.
✗ **D** 7: 7/8 is more than 3/4. One-fourth equals 2/8, so 3/4 = 6/8.

34. Answer: C **(NC.4.NF.1 · Identify equivalent)**

✗ **A** 4/10: 2/5 × 2/2 = 4/10, so this one is equivalent.

✗ **B** 6/15: 2/5 × 3/3 = 6/15, so this one is equivalent.

✓ **C** 4/15: The numerator was multiplied by 2 but the denominator by 3. They must be multiplied by the same number, so 4/15 is NOT equivalent.

✗ **D** 8/20: 2/5 × 4/4 = 8/20, so this one is equivalent.

35. Answer: C **(NC.4.NF.1 · Word problems)**

✗ **A** Maria: 6 is bigger than 3, but tenths are smaller pieces than fifths. The amounts are equal.

✗ **B** Ty: 3/5 = 6/10, so Ty did not eat more.

✓ **C** They ate the same amount: 3/5 × 2/2 = 6/10. Both ate the same amount.

✗ **D** There is not enough information: The pizzas are the same size, so you can compare the fractions.

36. Answer: B **(NC.4.NF.1 · Explain & prove)**

✗ **A** (3 + 2) / (4 + 2) = 5/6: Adding the same number changes the size of the fraction. 5/6 is more than 3/4.

✓ **B** (3 × 2) / (4 × 2) = 6/8: Multiplying top and bottom by the same number makes smaller pieces but keeps the same amount.

✗ **C** (3 × 2) / 4 = 6/4: Only the top was multiplied. 6/4 is more than one whole.

✗ **D** 3 / (4 × 2) = 3/8: Only the bottom was multiplied. 3/8 is less than 3/4.

37. Answer: A **(NC.4.NF.1 · Identify equivalent)**

✓ **A** 3/4: Divide the top and bottom by 3: 9 ÷ 3 = 3 and 12 ÷ 3 = 4.

✗ **B** 2/3: 2/3 = 8/12, not 9/12.

✗ **C** 9/10: Only the denominator changed. That doesn't make an equivalent fraction.

✗ **D** 1/3: 1/3 = 4/12, which is much less than 9/12.

38. Answer: B **(NC.4.NF.1 · Use fraction models)**

✗ **A** 1/2: 1/2 = 3/6, not 2/6.

✓ **B** 1/3: Divide the top and bottom by 2: 2/6 = 1/3. Both mark the same distance from 0.

✗ **C** 2/3: 2/3 = 4/6, twice as far from 0 as 2/6.

✗ **D** 1/6: 1/6 is only one sixth. 2/6 is two sixths.

39. Answer: A **(NC.4.NF.1 · Identify equivalent)**

✓ **A** 5/6 and 10/12: 5 × 2 = 10 and 6 × 2 = 12, so 5/6 = 10/12.

✗ **B** 3/8 and 4/9: 1 was added to the top and bottom. Adding doesn't make equivalent fractions.

✗ **C** 2/4 and 3/8: 2/4 = 4/8, not 3/8.

✗ **D** 1/5 and 2/5: The denominators are the same, but 2/5 has twice as many parts.

40. Answer: C **(NC.4.NF.1 · Find the missing number)**

✗ **A** 4: 4/100 is much smaller than 4/10. The numerator must grow too.

✗ **B** 14: 10 was added to the numerator. Multiply by 10 instead.

✓ **C** 40: 10 × 10 = 100, so 4 × 10 = 40.

✗ **D** 400: The numerator was multiplied by 100, but the denominator only by 10.

41. Answer: B **(NC.4.NF.1 · Word problems)**

✗ **A** 1: One scoop is only 1/8 cup, which is half of what she needs.

✓ **B** 2: 1/4 = 2/8, so she needs 2 scoops of 1/8 cup.

✗ **C** 4: 4 scoops = 4/8 cup = 1/2 cup. That's too much.

✗ **D** 8: 8 scoops = 8/8 = 1 whole cup. That's far too much.

42. Answer: B **(NC.4.NF.1 · Find the missing number)**

✗ **A** 3: 3/3 = 1 whole, but 12/16 is less than a whole.

✓ **B** 4: 3 × 4 = 12, so the denominator × 4 must be 16. 4 × 4 = 16, so 3/4 = 12/16.

✗ **C** 5: 5 × 4 = 20, not 16.

✗ **D** 8: 3/8 = 6/16, not 12/16.

43. Answer: D **(NC.4.NF.1 · Use fraction models)**

✗ **A** 2: 2/9 uses the numerator without changing it. That is much less than 2/3.
✗ **B** 3: 3/9 = 1/3, not 2/3.
✗ **C** 5: 5/9 is a little less than 2/3.
✓ **D** 6: 3 × 3 = 9, so 2 × 3 = 6. 2/3 = 6/9.

44. Answer: B **(NC.4.NF.1 · Explain & prove)**

✗ **A** He should have added 3: Adding any number to the top and bottom changes the fraction's size.
✓ **B** Adding does not keep the fractions equal; he must multiply both by the same number: Multiplying top and bottom by the same number keeps the amount the same. Adding does not.
✗ **C** Fractions can never be equivalent: Many fractions are equivalent, like 1/2 and 2/4.
✗ **D** He should have subtracted 2: Subtracting also changes the size of the fraction.

45. Answer: C **(NC.4.NF.1 · Identify equivalent)**

✗ **A** 1/8: 1/8 is only one of eight parts.
✗ **B** 7/8: 7/8 is one part short of a whole.
✓ **C** 8/8: When the numerator equals the denominator, every part is shaded. 8/8 = 1.
✗ **D** 8/1: 8/1 means 8 wholes, not 1.

46. Answer: B **(NC.4.NF.1 · Word problems)**

✗ **A** 1/2: 1/2 = 6/12. The corn covers 8 rows, not 6.
✓ **B** 2/3: Divide the top and bottom by 4: 8/12 = 2/3.
✗ **C** 3/4: 3/4 = 9/12, one row more than the corn.
✗ **D** 4/5: Fifths don't divide 12 rows evenly, and 4/5 is not equal to 8/12.

47. Answer: A **(NC.4.NF.1 · Identify equivalent)**

✓ **A** 2/6, 3/9, 4/12: Each one is 1/3 times 2/2, 3/3, or 4/4.
✗ **B** 2/3, 3/4, 4/5: These grow bigger each time. None equals 1/3.
✗ **C** 1/6, 1/9, 1/12: Only the denominator changed, so each piece gets smaller.
✗ **D** 3/1, 6/2, 9/3: These all equal 3 wholes, not 1/3.

48. Answer: B **(NC.4.NF.1 · Use fraction models)**

✗ **A** 1/5: 1/5 = 2/10, not 5/10.
✓ **B** 1/2: Divide the top and bottom by 5: 5/10 = 1/2. Half the bar is shaded.
✗ **C** 5/5: 5/5 is a whole bar, but only half is shaded.
✗ **D** 2/5: 2/5 = 4/10, not 5/10.

49. Answer: B **(NC.4.NF.1 · Find the missing number)**

✗ **A** 6: 6/24 = 2/8. Multiply 3 by 3, not by 2.
✓ **B** 9: 8 × 3 = 24, so 3 × 3 = 9.
✗ **C** 12: 12/24 = 1/2, which is more than 3/8.
✗ **D** 16: 16 was added to the denominator. Multiply instead.

50. Answer: C **(NC.4.NF.1 · Word problems)**

✗ **A** Lena read more: 2/4 = 1/2 and 5/10 = 1/2. Lena didn't read more.
✗ **B** Max read more: 5 is bigger than 2, but tenths are smaller pieces. The amounts are equal.
✓ **C** They both read 1/2 of the book: 2/4 = 1/2 and 5/10 = 1/2. They read the same amount.
✗ **D** They both read 1/4 of the book: Both fractions equal 1/2, not 1/4.

51. Answer: C **(NC.4.NF.1 · Find the missing number)**

✗ **A** 5: 5/16 is half of 5/8. The numerator must be multiplied too.
✗ **B** 8: 8/16 = 1/2, which is less than 5/8.
✓ **C** 10: 8 × 2 = 16, so 5 × 2 = 10.
✗ **D** 13: 8 was added to the numerator. Multiply by 2 instead.

52. Answer: D **(NC.4.NF.1 · Identify equivalent)**

✗ **A** 1/2: 1/2 = 6/12, not 10/12.
✗ **B** 2/3: 2/3 = 8/12, not 10/12.
✗ **C** 3/4: 3/4 = 9/12, not 10/12.
✓ **D** 5/6: Divide the top and bottom by 2: 10/12 = 5/6.

53. Answer: B **(NC.4.NF.1 · Find the missing number)**

✗ **A** 5: 5/100 = 1/20, which is too small.
✓ **B** 20: 5 × 20 = 100, so 1 × 20 = 20. 1/5 = 20/100.
✗ **C** 50: 50/100 = 1/2, which is more than 1/5.
✗ **D** 95: This subtracts 5 from 100. Multiply instead.

54. Answer: A **(NC.4.NF.1 · Identify equivalent)**

✓ **A** 2/3 = 4/6: 2 × 2 = 4 and 3 × 2 = 6, so these are equal.
✗ **B** 1/4 = 2/6: 1/4 = 2/8, not 2/6.
✗ **C** 3/5 = 5/7: 2 was added to the top and bottom, which changes the amount.
✗ **D** 4/8 = 1/4: 4/8 = 1/2, not 1/4.

55. Answer: B **(NC.4.NF.1 · Use fraction models)**

✗ **A** 1: 1/3 = 2/6, half of what is shaded on top.
✓ **B** 2: Divide the top and bottom by 2: 4/6 = 2/3.
✗ **C** 3: 3/3 is the whole bar, but the top bar isn't fully shaded.
✗ **D** 4: There are only 3 parts in the bottom bar, so you can't shade 4.

56. Answer: A **(NC.4.NF.1 · Explain & prove)**

✓ **A** Yes, because 3 is half of 6 and 4 is half of 8: When the numerator is half the denominator, the fraction equals 1/2.
✗ **B** No, because 4/8 has bigger numbers: Bigger numbers don't mean a bigger amount. Eighths are smaller pieces than sixths.
✗ **C** No, because the denominators are different: Fractions with different denominators can still be equivalent.
✗ **D** Only 3/6 equals 1/2: 4/8 also equals 1/2, because 4 is half of 8.

57. Answer: C **(NC.4.NF.1 · Find the missing number)**

✗ **A** 3: 5/3 is more than one whole, but 15/18 is less than one whole.
✗ **B** 5: 5/5 = 1 whole, but 15/18 is less than one whole.
✓ **C** 6: 5 × 3 = 15, so the denominator × 3 must be 18. 6 × 3 = 18.
✗ **D** 8: 10 was subtracted from 18. Divide by 3 instead.

58. Answer: D **(NC.4.NF.1 · Word problems)**

✗ **A** 2/5: 2/5 = 4/10, not 8/10.
✗ **B** 3/5: 3/5 = 6/10, not 8/10.
✗ **C** 8/5: Only the denominator changed. 8/5 is more than one whole.
✓ **D** 4/5: Divide the top and bottom by 2: 8/10 = 4/5.

59. Answer: B **(NC.4.NF.1 · Explain & prove)**

✗ **A** 2/5 + 4/10 = 6/15: Adding fractions doesn't prove two fractions are equivalent.
✓ **B** (2 × 3) / (5 × 3) = 6/15: Multiplying the top and bottom by 3 gives 6/15, so the fractions are equal.
✗ **C** (2 + 4) / (5 + 10) = 6/15: Adding different numbers to the top and bottom doesn't keep the fraction the same.
✗ **D** (2 × 3) / 5 = 6/15: Only the top was multiplied, which gives 6/5, not 6/15.

60. Answer: A **(NC.4.NF.1 · Identify equivalent)**

✓ **A** 1/4: Divide the top and bottom by 3: 3/12 = 1/4.
✗ **B** 1/3: 1/3 = 4/12, not 3/12.
✗ **C** 3/4: 3/4 = 9/12, three times as much.
✗ **D** 1/12: 1/12 is only one of twelve parts.